

Scenario-Based Questions and Model Identification Techniques

Scenario 1: Waterfall Model

Scenario:

A government agency plans to develop a tax management system. The requirements have been thoroughly documented and approved before development begins. Regulatory policies are stable, and no major changes are expected during the project. The agency requires detailed documentation for each development phase.

Question:

Which software process model is most suitable? Justify your answer.

Answer:

The **Waterfall Model** is the most suitable because the requirements are clearly defined and stable. The project can proceed sequentially through requirements analysis, design, implementation, testing, deployment, and maintenance without significant modifications.

Identification Clues:

- Requirements are fixed.
- No expected changes.
- Sequential execution.
- Strong documentation requirements.

Exam Trick:

If you see phrases such as *well-defined requirements*, *stable requirements*, *no changes expected*, or *linear development*, choose **Waterfall**.

Scenario 2: V-Model

Scenario:

A healthcare company is developing software for an insulin pump. Since human safety is involved, every development phase must be accompanied by a corresponding testing phase to ensure reliability and compliance.

Question:

Which software process model should be selected?

Answer:

The **V-Model** should be selected because it integrates verification and validation activities throughout development. Each development stage has a corresponding testing stage.

Identification Clues:

- Safety-critical software.
- Verification and validation.
- Extensive testing requirements.

- Strong quality assurance focus.

Exam Trick:

Whenever you see *testing, verification, validation, quality assurance, or safety-critical systems*, think **V-Model**.

Scenario 3: Prototyping Model

Scenario:

A startup wants to develop a social networking application. Potential users have only a general idea of what features they want and continue changing their opinions after viewing sample interfaces.

Question:

Which software process model is most appropriate?

Answer:

The **Prototyping Model** is most appropriate because customer requirements are unclear and evolving. Building prototypes allows users to visualize the system and refine their requirements.

Identification Clues:

- Requirements are unclear.
- Customers are uncertain.
- Frequent requirement changes.
- User feedback is essential.

Exam Trick:

Keywords such as *unclear requirements, customer confusion, feedback, mock-up, or prototype* indicate the **Prototyping Model**.

Scenario 4: Incremental Development

Scenario:

An online learning platform plans to release its system in stages. The first release will contain student registration, the second release will add course management, and the third release will include online examinations.

Question:

Which software process model should be used?

Answer:

The **Incremental Development Model** should be used because the system is delivered in multiple increments, each providing additional functionality.

Identification Clues:

- Feature-by-feature delivery.
- Multiple releases.
- Prioritized requirements.
- Continuous enhancement.

Exam Trick:

If the scenario mentions *phases, modules, releases, increments, or feature-by-feature development*, select **Incremental Development**.

Scenario 5: RAD Model

Scenario:

A retail company requires an inventory management system within three months. Users are available for continuous interaction, and the organization has experienced developers capable of parallel development.

Question:

Which software process model is most suitable?

Answer:

The **RAD (Rapid Application Development) Model** is most suitable because it emphasizes rapid development, user involvement, and quick delivery through parallel development activities.

Identification Clues:

- Short deadline.
- Fast delivery required.
- User participation.
- Parallel development.

Exam Trick:

Keywords such as *90 days*, *rapid delivery*, *time-boxed*, *quick development*, or *parallel development* indicate **RAD**.

Scenario 6: Rational Unified Process (RUP)

Scenario:

A multinational corporation is developing an enterprise resource planning (ERP) system involving finance, human resources, inventory, and logistics. The project contains significant technical risks and evolving business requirements.

Question:

Which software process model is most appropriate?

Answer:

The **Rational Unified Process (RUP)** is most appropriate because it is iterative, risk-driven, and supports continuous refinement of requirements.

Identification Clues:

- Large-scale project.
- High risk.
- Iterative development.
- Continuous stakeholder involvement.

Exam Trick:

Whenever you see *risk management*, *large project*, *enterprise system*, *iterations*, or *continuous refinement*, choose **RUP**.

Golden Rule for Model Identification

Scenario Keyword	Correct Model
Stable and fixed requirements	Waterfall
Testing, verification, validation	V-Model
Unclear requirements, customer feedback	Prototyping
Feature-by-feature delivery	Incremental Development
Fast development, 90-day deadline	RAD
Large project, high risk, iterative process	RUP

One-Line Shortcut for Exams

- Stable Requirements $\Rightarrow$ Waterfall
- Stable Requirements + Heavy Testing $\Rightarrow$ V-Model
- Unclear Requirements $\Rightarrow$ Prototyping
- Multiple Releases $\Rightarrow$ Incremental Development
- Fast Delivery + User Feedback $\Rightarrow$ RAD
- Large Project + High Risk + Iteration $\Rightarrow$ RUP

Software Process Model Identification Tips and Tricks

Model	Keywords in Scenario	When to Use	Exam Trick
Waterfall	Stable requirements, fixed requirements, detailed documentation, no changes expected, sequential phases	When all requirements are clearly known before development starts and changes are unlikely	If the scenario says " <i>requirements are well-defined</i> " or " <i>no changes expected</i> ", immediately think Waterfall .
V-Model	Verification, validation, testing, quality assurance, defect detection, safety-critical system	When software quality and testing are extremely important	If you see " <i>every phase has a corresponding testing phase</i> " or " <i>extensive testing required</i> ", choose V-Model .
Prototyping	Unclear requirements, customer feedback, evolving requirements, user interaction, mock-up	When customers cannot clearly define their requirements	If users are confused about requirements or keep changing their minds, select Prototyping .
Incremental Development	Modules, releases, increments, phases, feature-by-feature delivery, prioritized requirements	When software can be delivered in small functional parts	If the scenario mentions " <i>first release</i> ", " <i>second release</i> ", or " <i>module-wise delivery</i> ", choose Incremental Development .
RAD	Rapid development, 90 days, time-boxed, parallel development, quick delivery, user involvement	When software must be developed quickly with continuous customer participation	If you see " <i>very short deadline</i> " or " <i>rapid delivery required</i> ", think RAD .
RUP	Large project, high risk, iterative development, stakeholder collaboration, continuous refinement	When developing complex enterprise systems with significant risks	If the scenario mentions " <i>high risk</i> " or " <i>iterative development</i> ", choose RUP .

5-Second Exam Formula

Stable Requirements	$\Rightarrow$ Waterfall
Stable + Heavy Testing	$\Rightarrow$ V-Model
Unclear Requirements	$\Rightarrow$ Prototyping
Multiple Releases	$\Rightarrow$ Incremental Development
Fast Delivery + User Feedback	$\Rightarrow$ RAD
Large Project + High Risk + Iteration	$\Rightarrow$ RUP